

Module 04: Feature Engineering II - Group-Level Transforms and Contextual Modeling

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Abstract

Numbers in isolation are often meaningless. If a column says a user spent \$150, is that high or low? Without context, the model cannot know. This note covers the use of the `.transform()` function to build contextual features that compare individual data points against their peer groups, transforming raw numbers into behavioral signals.

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1 The Concept of Contextual Modeling

Machine learning models struggle to understand "normality." To help them, we must explicitly calculate what is normal for a specific group and then show the model how an individual row deviates from that norm.

2 The .transform() Function

Unlike .agg(), which reduces the number of rows to a summary table, .transform() calculates a statistic for a group but **broadcasts it back** to the original shape of the DataFrame.

Essential Syntax

```
df['group_mean'] = df.groupby('Group_Col')['Target_Col'].transform('mean')
```

3 Practical Contextual Features

3.1 Relative Deviation

Is this user spending more than their personal average?

$$\text{Diff.From.Avg} = \text{Current.Spend} - \text{User.Avg.Spend} \quad (1)$$

```
1 df['user_avg'] = df.groupby('User_ID')['Spend'].transform('mean')
2 df['spend_vs_user_avg'] = df['Spend'] - df['user_avg']
3
```

3.2 Group-Based Z-Scores

Is this house priced abnormally high for its neighborhood?

$$z_{group} = \frac{x - \mu_{group}}{\sigma_{group}} \quad (2)$$

```
1 df['hood_avg'] = df.groupby('Neighborhood')['Price'].transform('mean')
2 df['hood_std'] = df.groupby('Neighborhood')['Price'].transform('std')
3 df['price_z_hood'] = (df['Price'] - df['hood_avg']) / df['hood_std']
4
```

4 Practice Problems

P1: When should you use .transform() instead of .merge()?

Answer: Use .transform() for simple statistical broadcasts (mean, sum, std) because it is faster and requires less code. Use .merge() when you need to join complex summary tables with multiple different metrics.

P2: You are predicting fraud in transactions. Why is Amount - Avg_Amount_For_That_Card a better feature than just Amount?

Answer: Because \$500 might be normal for a wealthy user but highly suspicious for a student. The relative difference provides the "behavioral context" that raw numbers lack.